Class08: Breast Cancer Mini Project

Qiman A17912409

Background

The goal of this mini-project is for you to explorea complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from the needle aspiration(FNA)

Data import

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)
```

Examine the input data the ensure column names are set correctly

```
head(wisc.df,4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
```

```
842302          0.2419               0.07871   1.0950    0.9053        8.589
842517          0.1812               0.05667   0.5435    0.7339        3.398
84300903        0.2069               0.05999   0.7456    0.7869        4.585
84348301        0.2597               0.09744   0.4956    1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

Make sure we don't accidentally include diagnosis column in our analysis.

```r
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
```

Now we set up a new vector called diagnosis and store this as a factor for later.

```r
# Create diagnosis vector for later
diagnosis <-  as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

> Q1.How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

There are 569 observations in this dataset.

> Q2.How many of the observations have a malignant diagnosis?

```r
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

There are 212 observations with a malignant diagnosis.

> Q3. How many variables/features in the data are suffixed with _mean?

We can use the `grep()` function to help us here:

```r
length(grep("_mean",colnames(wisc.data), value=TRUE))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```r
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

The proportion of the original variance is .4427 so 44.275 of the total variance in the dataset is captured by PC1.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

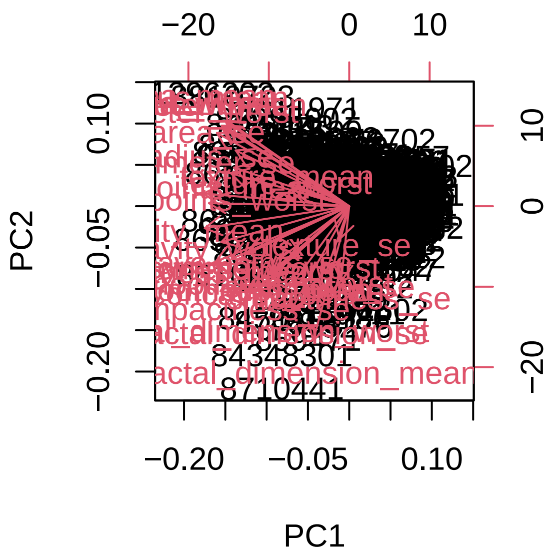
You need 3 principal components to describe at least 70% of the original variance in the data.

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

You need 7 principal components to describe at least 70% of the original variance in the data.

Creating a biplot:

```
biplot(wisc.pr)
```



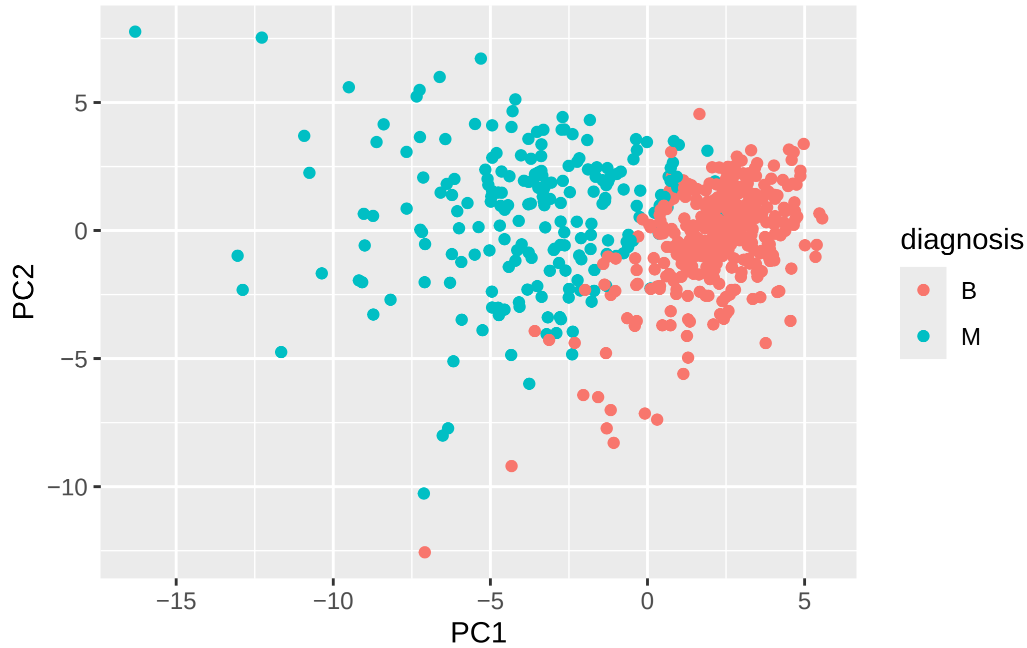
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

This plot is a huge mess and it is very difficult to understand. All the values are highly clustered and it is impossible to read the plot properly.

Let's see the "PC score plot"

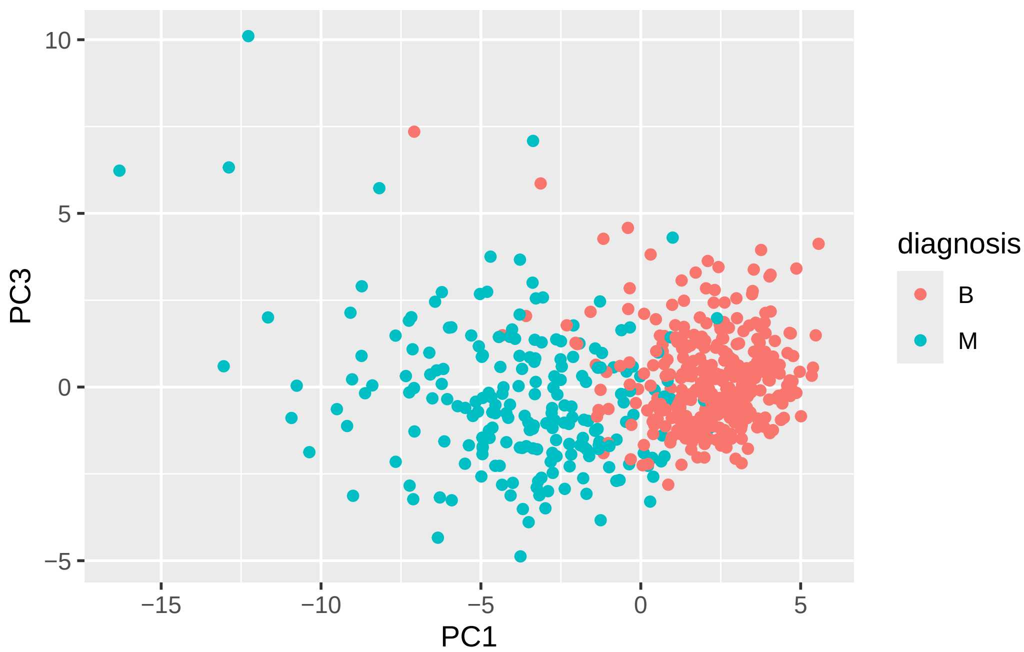
```
library(ggplot2)

ggplot(wisc.pr$x)+
  aes(PC1,PC2, col=diagnosis)+
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x)+
  aes(PC1,PC3, col=diagnosis)+
  geom_point()
```



Overall the plots show that principal component 1 is capturing a separation of malignant and benign samples.

Variance Explained

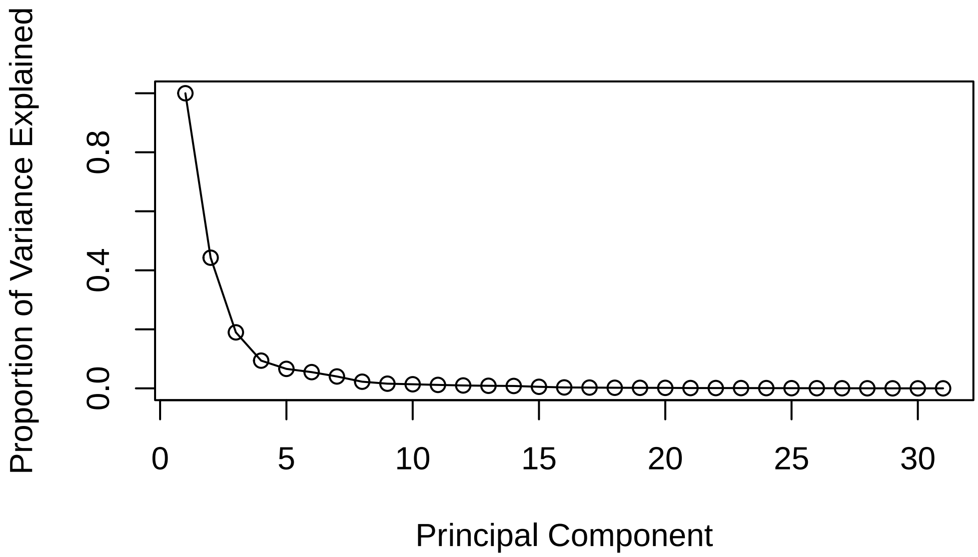
Use a scree plot to show how much variance each PC captures. Look for an "elbow".

```
pr.var <- wisc.pr$sdev^2
  head(pr.var)
```

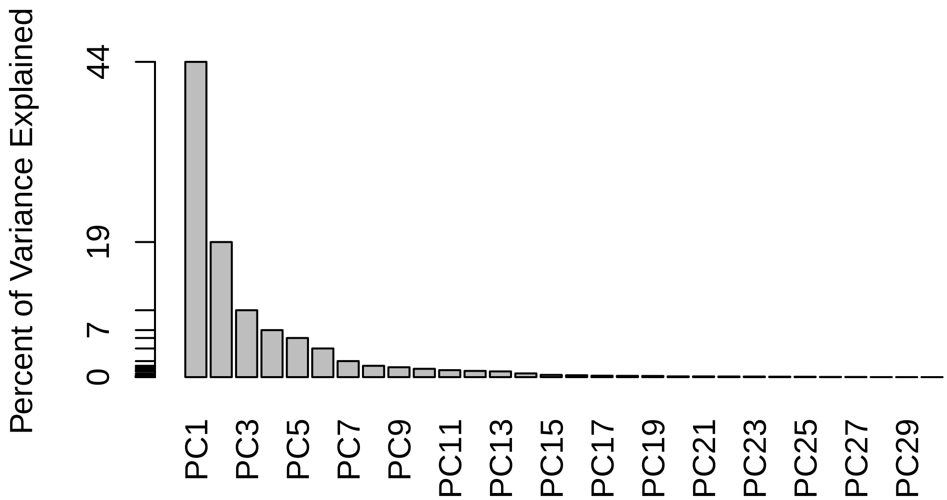
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
#Variance explained by each principal component: pve
pve <- pr.var/sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA results

> Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
# Loading for concave.points_mean on PC1
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
# Check largest absolute loadings for PC1
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

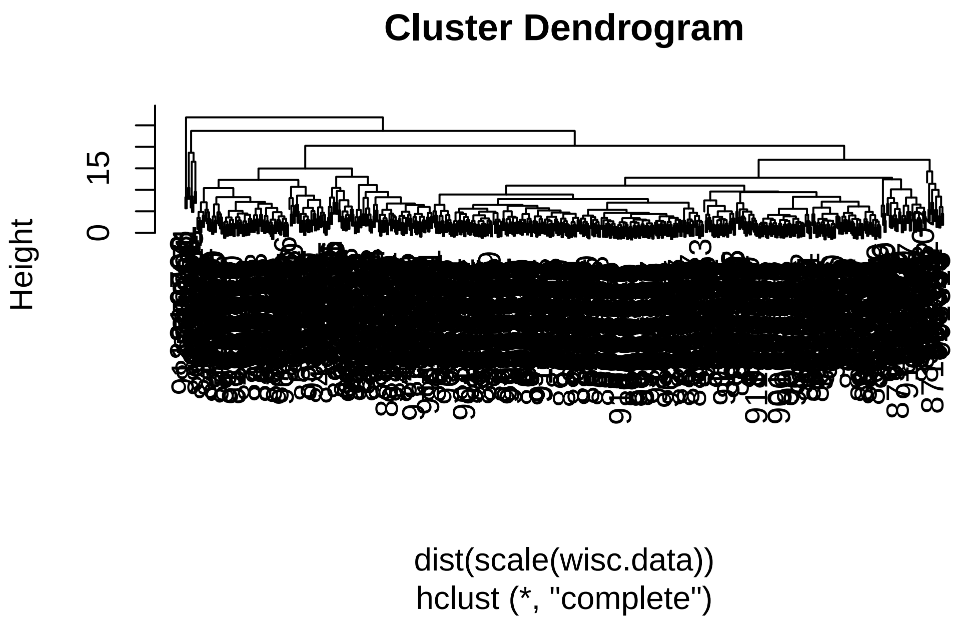
```
concave.points_mean          concavity_mean    concave.points_worst
         0.26085376              0.25840048              0.25088597
   compactness_mean         perimeter_worst         concavity_worst
         0.23928535              0.23663968              0.22876753
       radius_worst          perimeter_mean              area_worst
         0.22799663              0.22753729              0.22487053
```

```
          area_mean            radius_mean           perimeter_se
         0.22099499             0.21890244             0.21132592
  compactness_worst              radius_se                area_se
         0.21009588             0.20597878             0.20286964
   concave.points_se         compactness_se           concavity_se
         0.18341740             0.17039345             0.15358979
    smoothness_mean          symmetry_mean fractal_dimension_worst
         0.14258969             0.13816696             0.13178394
   smoothness_worst         symmetry_worst          texture_worst
         0.12795256             0.12290456             0.10446933
       texture_mean   fractal_dimension_se fractal_dimension_mean
         0.10372458             0.10256832             0.06436335
        symmetry_se             texture_se          smoothness_se
         0.04249842             0.01742803             0.01453145
```

For the first principal component the loading for concave.points_mean is approximately -0.2609 meaning that concave.points_mean contributes strongly to PC1. There are no features with larger contributions than this one.

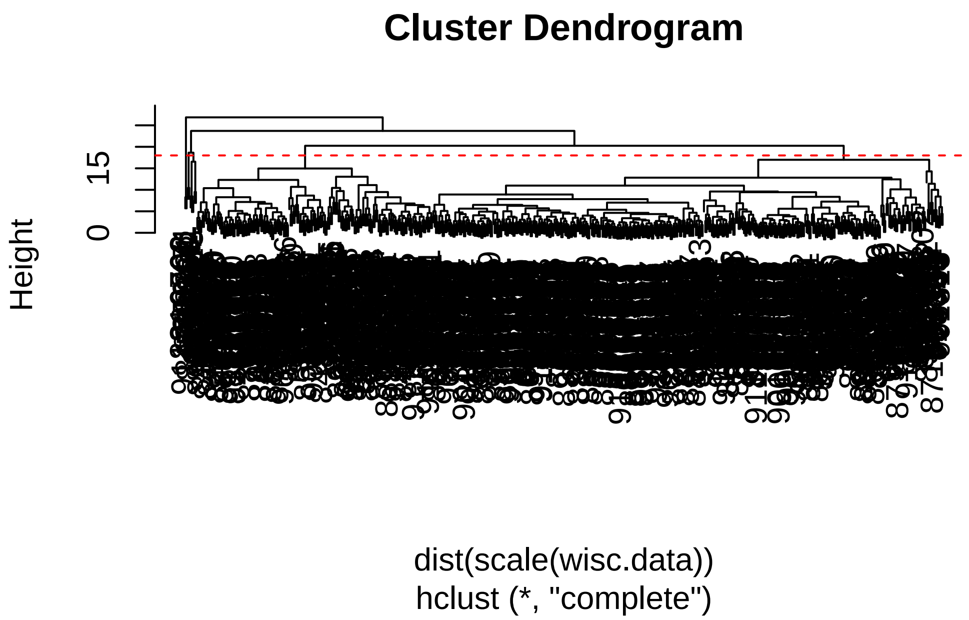
Hierarchical Clustering

```r
wisc.hclust <- hclust( dist( scale(wisc.data)))
plot(wisc.hclust)
```



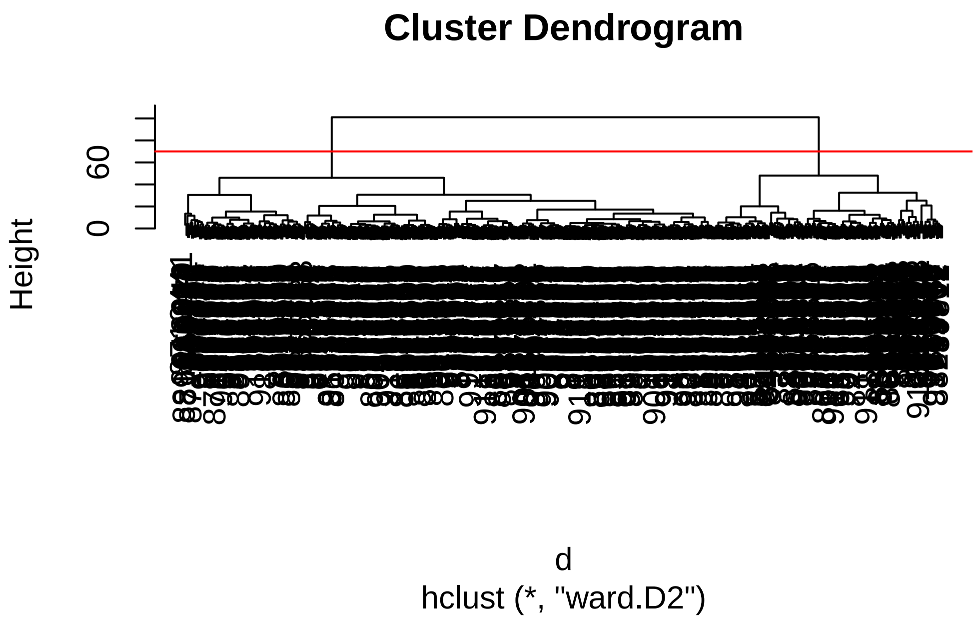
Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)
abline(h=18, col="red", lty=2)
```



at h=18 the clustsering model has 4 clusters. ##Combining methods

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d,method="ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col="red")
```




```r
wisc.hclust.clusters <- cutree(wisc.hclust, h=20)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

ward.D2 gives the best results because it produce the clearest and most interpretable cluster structure. Te clusters were more compact and distinct.

```r
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

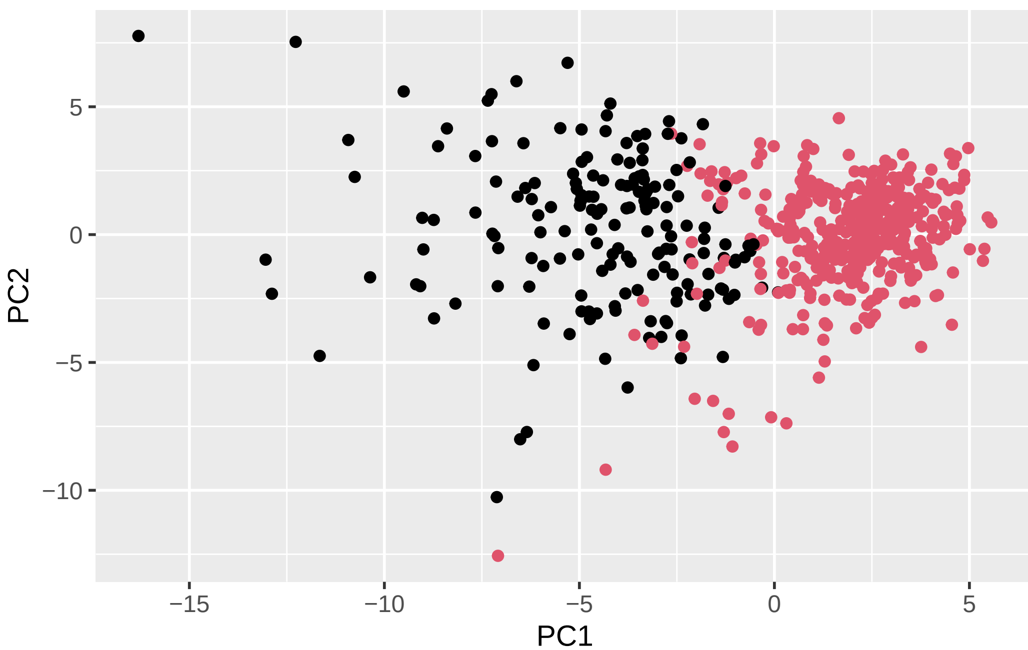
```
grps
  1   2
171 398
```

How does this clustering `grps` correspond to the expert `diagnosis`

```r
table(diagnosis, grps)
```

```
         grps
diagnosis   1   2
        B   6 351
        M 165  47
```

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```




```r
l <- dist(wisc.pr$x[,1:7])
wisc.pr.hclust1 <- hclust(l,method="ward.D2")

wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust1, k=2)
```

> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

```r
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

The two cluster model separates the diagnoses well but not perfectly. Cluster 1 is mostly malignant with 28 benign. Cluster two is mostly benign with 24 malignant. The model does distinguish between the two groups fairly well with the accuracy being high.

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```r
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```r
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

The hierarchical clustering model without PCA can separate benign and malignant diagnoses but the clusters are more fragmented and less interpretable. For example small clusteres 2 and 4 contain few samples, there is some separation but it is fragmented intwo 4 clusters. The PCA-based clustering performs better overall giving a clearer separation between the two groups. One cluster is mostly malignant while the other is mostly benign. This is a much cleaner 2-group separation.

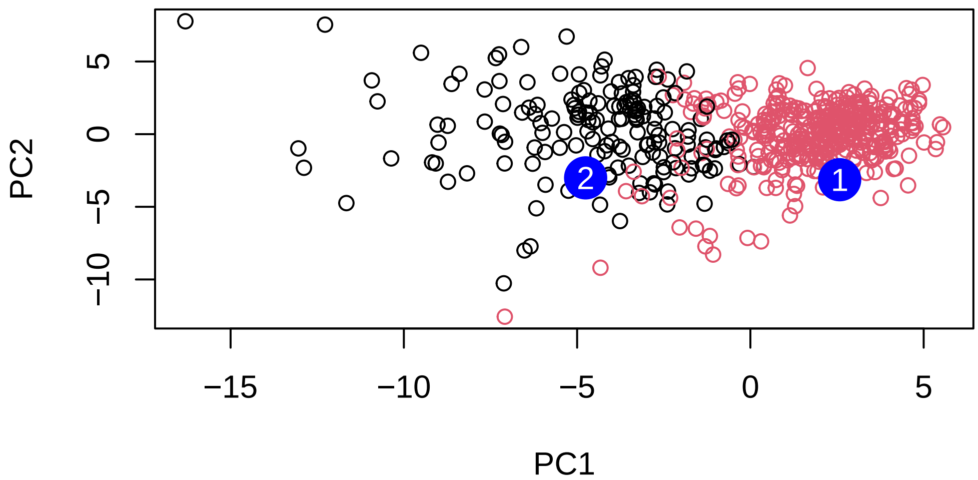
Prediction

We will use the `predict()` function that will take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



> Q16. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize patient 2 because it has PC1=-4.25 which places it closer to the cluster of malignant samples. As seen in the plot patient 2 clusters closer to the malignant tumor samples.